Neuron Architecture

Hybrid execution model for the workflow engine: plans are generated and orchestrated server-side, but task execution can be routed to external processes ("neurons") — including desktop clients holding user-specific MCP connections — or to in-cluster workers, based on declared capabilities and scope.

Motivation

The baseline architecture (see `ARCHITECTURE.md`) executes all tasks in server-side worker processes. This works for generic compute (scripts, HTTP, LLM calls) but breaks for user-specific integrations:

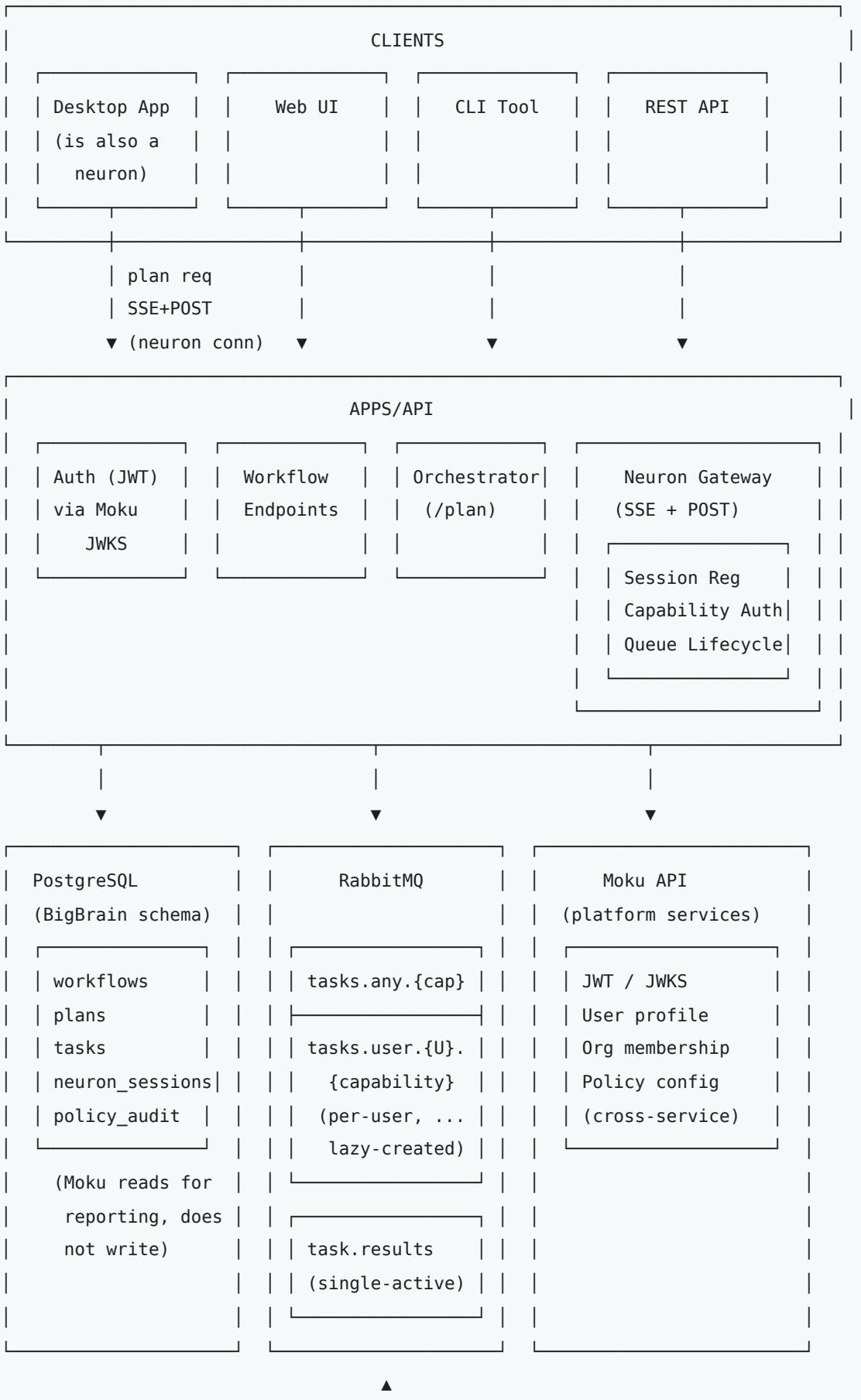
- **MCP connections are per-user.** A desktop client authenticated to Google Drive as user `abc` cannot legally (or safely) process a Drive task on behalf of user `xyz`.
- **Capability availability varies by client.** A desktop binary ships with different MCP servers than another; a web client has none.
- **Trust boundaries differ.** A server-side worker has database credentials and service-account trust; a user-installed desktop neuron does not.

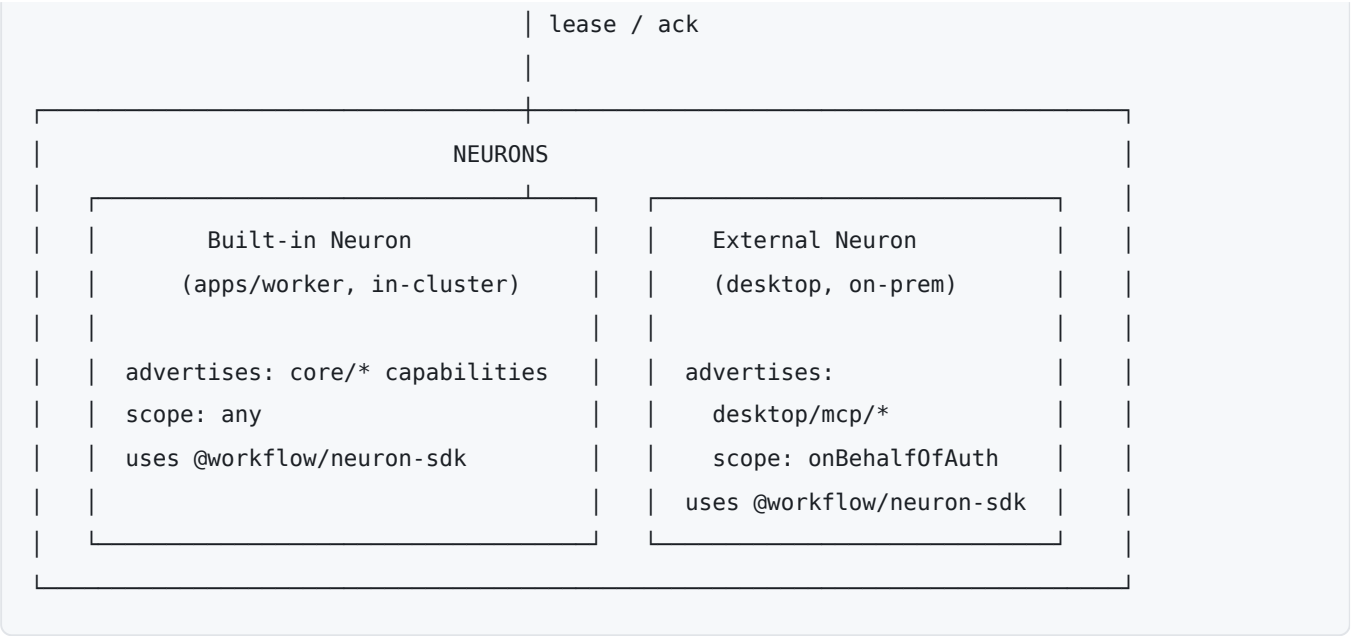
The neuron architecture introduces a routing layer that matches tasks to executors by capability, scope, and authorization, while keeping plan generation and orchestration centralized.

Core Concepts

- **Neuron** — any process that executes tasks. Identified by an authenticated identity (JWT from Moku), advertises one or more capabilities at registration time. The built-in `apps/worker` is itself a neuron; external neurons (desktop, per-tenant agents) connect over the wire protocol.
- **Capability** — a typed function signature that a neuron can execute. Identified by a namespaced type string (`core/script.python`, `desktop/mcp/drive.read`). Schemas for input/output are not global; they travel with the plan (see *Plan-Time vs Registration-Time Schemas*).
- **Scope** — who a capability advertisement is valid for. Either `any` (any user's task) or `{ onBehalfOf: userId }` (only that user's tasks). Scope is enforced by the gateway at both registration time and routing time.
- **Gateway** — the server-side component that accepts neuron connections, authorizes capability advertisements, binds consumer channels to per-user queues, and forwards work. Mounted as routes on `apps/api` for v1, isolated enough to extract into its own service if scale demands.
- **On-behalf-of** — the user identity a task executes for. Set at plan time (from the workflow's owner), stamped on every task, used to select the target queue.

System Diagram





Execution Model

Registration

A neuron opens an SSE stream to `/neuron/events` (server → neuron) and sends a `register` frame via `POST /neuron/register` with its JWT in the `Authorization` header. The frame carries:

- `neuronId` — stable id per device/process
- `capabilities[]` — each with `type`, `scope`, `concurrency`

The gateway:

1. Verifies the JWT via cached Moku JWKS.
2. For each advertised capability, checks the **namespace authorization table** — which identities are permitted to advertise which namespace prefixes.
3. Invokes `PolicyEngine.checkRegistration()` (see *Policy Integration*).
4. For user-scoped advertisements, verifies the auth token is authorized for that user.
5. Lazily declares the matching RabbitMQ queues and opens consumer channels with `prefetch = concurrency`.
6. Persists the session to `neuron_sessions` for UI visibility and routing lookups.

Registration is the **routing concern** only. No schemas are exchanged at registration time — schemas live with each plan.

Capability Scoping

Scope is a property of each capability advertisement, not the neuron as a whole. One neuron may advertise mixed-scope capabilities:

```
{
  "capabilities": [
    { "type": "desktop/mcp/drive.read", "scope": { "onBehalfOf": "user:abc" }, "concurrency":
    { "type": "desktop/mcp/slack.post", "scope": { "onBehalfOf": "user:abc" }, "concurrency":
    { "type": "core/script.python",     "scope": "any",                        "concurrency":
  ]
}
```

The auth token's user binding is the **only** authority for `onBehalfOf` scope — a neuron cannot advertise a scope it is not authorized for; registration is rejected.

Per-User Queues

Queues for user-scoped capabilities are **lazily declared per (user, capability)** pair:

- Name: `tasks.user.{userId}.{capabilityType}` (e.g., `tasks.user.abc.desktop.mcp.drive.read`)
- `durable: true`
- `x-expires: 7d` — broker auto-deletes queue after 7d with no consumers and no messages
- `x-message-ttl: 24h` — individual tasks expire if unclaimed past this window
- DLX → `tasks.expired` — exposes timed-out tasks to the planner for re-plan or fail decisions

Queues for scope-`any` capabilities use a shared queue per capability type: `tasks.any.{capabilityType}`.

Leasing and Restart Semantics

Tasks are leased, not dequeued. A neuron consumes a message without acking; the gateway writes the task's DB row to `assigned` with a lease expiry. The ack is sent only on terminal success. On disconnect, the broker redelivers the message automatically; a reaper flips stale `assigned` rows back to `pending` and the task resumes on the next consumer (possibly the same user's different device).

Consequence for handler authors: handlers must be idempotent, or use capability-native idempotency keys (e.g., Google Drive `requestId`). Restart semantics mean "a task that completed a side effect but failed to ack will re-run." This is a per-handler contract, not a framework-level guarantee.

Concurrency

Each capability advertisement carries `concurrency`. The gateway translates this into AMQP `basic.qos` `prefetch_count` on the consumer channel, so the broker handles backpressure — once the neuron has N unacked messages, it stops receiving new ones until acks flow.

Neurons may send `UPDATE_CAPABILITY` frames to change concurrency mid-session (e.g., desktop plugged in); the gateway applies new prefetch on next delivery without disturbing in-flight leases.

Heartbeats

Every ~10s, the neuron sends:

```
{ "type": "heartbeat", "neuronId": "...", "inFlight": [
    { "leaseId": "...", "taskId": "...", "startedAt": "..." }, ...
]}
```

Heartbeats (a) renew all active lease expiries in one round-trip, (b) let the gateway detect wedged neurons (inFlight full + no progress), (c) power the UI's "3/4 busy" display.

Plan-Time vs Registration-Time Schemas

Capabilities are named routing addresses; **their schemas are not globally registered**. Instead:

- The client initiating a workflow owns the handler catalog — the set of capability types available in its context, each with input/output schemas (zod → JSON Schema). The desktop knows its MCP tools; the server knows its bundled handlers.
- At plan creation, the client posts the catalog alongside the goal:

```
POST /plan
{
  "goal": "summarize the sales deck in Drive and post to #launches",
  "onBehalfOf": "user:abc",
  "handlers": [
    { "type": "desktop/mcp/drive.read", "input": {...}, "output": {...} },
    { "type": "core/llm.summarize",    "input": {...}, "output": {...} },
    { "type": "desktop/mcp/slack.post",  "input": {...}, "output": {...} }
  ]
}
```

- The LLM planner receives the catalog in its prompt context and emits a typed plan (dot-path bindings from context into each step's input).
- The planner validates its own plan against the catalog before persisting (type-compatibility check on every binding).
- The catalog is **snapshotted with the plan** — replay, debugging, re-execution use the schemas that were valid at plan time, not whatever is current.

Execution-time validation happens in the neuron SDK (input-schema check before invoking the handler, output-schema check before acking). Schema mismatches → specific nack codes; the orchestrator surfaces them to the user with a clear "plan expected shape X, neuron produced shape Y" error.

Capability Namespacing and Authorization

Capability types are namespaced: `{owner}/{package}/{name}`. Namespaces are not cosmetic — they are the primary key on the **namespace authorization table** that governs which identities are permitted to advertise which capabilities.

Namespace	Permitted advertisers	Typical scope
`core/*`	Server service identities (built-in worker)	`any`
`desktop/*`	Signed/attested desktop binary identities	`onBehalfOfAuth`
`org/{orgId}/*`	Tenant-installed extensions	`any` within org
`user/{userId}/*`	Personal extensions	`onBehalfOfAuth` for that user

Without enforced namespace authorization, capability collision leads to silent mis-routing or capability squatting (a malicious neuron claiming `core/*` addresses and siphoning tasks). The gateway rejects any registration where the advertiser's identity is not authorized for the namespace — collisions become visible failures at registration, not silent routing failures at runtime.

Wire Protocol

Transport: **SSE for server → neuron + HTTPS POST for neuron → server**, with correlation ids linking responses. Enterprise-friendly (traverses proxies cleanly, looks like normal HTTPS), matches the transport style of OpenAI/Anthropic streaming APIs and MCP's HTTP surface.

Message Types (overview — full schema in `packages/neuron-protocol`)

Neuron → Server (POST):

- `register` — open session with capabilities
- `update-capability` — change concurrency or scope for an existing capability
- `heartbeat` — lease renewal + in-flight state
- `ack` — task completed successfully, with output
- `nack` — task failed (retryable | terminal) with reason
- `progress` — in-progress status update for UI

Server → Neuron (SSE):

- `registered` — registration confirmed, includes effective capability bindings after policy/auth filters
- `lease` — a task to execute (`leaseId` , task payload with input schema)
- `cancel` — cancel a specific lease (triggered by workflow cancellation)
- `capability-revoked` — a capability was deauthorized mid-session; in-flight leases on that capability must fail
- `disconnect` — graceful close from server

SSE resumption uses `Last-Event-ID` for short reconnects; longer disconnects go through a fresh `register` (all prior leases already redelivered by the broker).

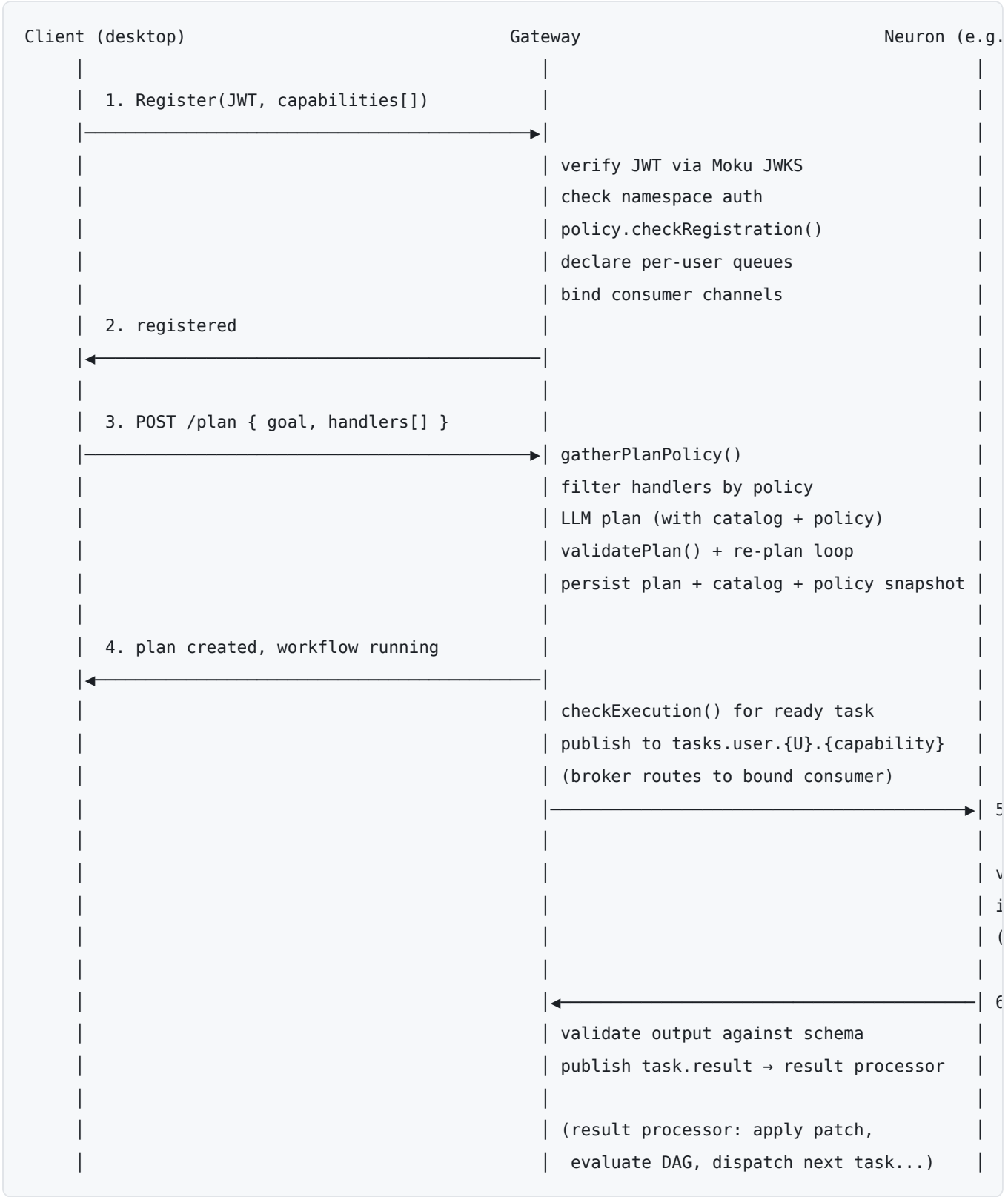
Policy Integration

Four hook points, each at a distinct lifecycle stage. Initially implemented as a `NoOpPolicyEngine` that always allows; real engine design deferred. Interface shipped in `packages/policy`:

Hook	Call Site	Input	Failure Mode
`checkRegistration`	Gateway, in register handler	identity, capability, scope	Registration rejected with code
`gatherPlanPolicy`	Planner, start of plan pipeline	goal, onBehalfOf, handlers	Opaque `PolicyContext` returned (NoOp → undefined)
`validatePlan`	Orchestrator, after LLM plan emission	plan, policyContext	Re-plan with violations, bounded retries, else terminal fail
`checkExecution`	Dispatcher, immediately before publish	task, plan, policyContext	Task fails with "policy changed since plan creation"

The `PolicyContext` returned by `gatherPlanPolicy` is **persisted alongside the plan**. `checkExecution` uses it as the at-plan-time snapshot for divergence detection — errors can name the specific policy that changed between plan time and execution time.

Data Flow: Hybrid Plan Execution



Moku Boundary (Data Ownership)

BigBrain owns its own Postgres **schema** (not a separate physical database). Moku retains read access for reporting and is the schema authority (migrations reviewed through Moku's coordination process); no other service writes to BigBrain's tables. This solves the cross-service coordination problem at the source.

BigBrain ↔ Moku API consumption surface:

Capability	Cadence	Cached
JWT verification (JWKS fetch)	Rare (key rotation)	Yes, in-memory
User profile lookup	On-demand (UI, audit)	Yes, short TTL
Org membership	On auth	Yes, per-session
Policy config	On plan start	Yes, by version hash

Hot-path operations — task state transitions, lease renewals, heartbeats, context patches — hit the BigBrain schema directly. Moku is not in the hot path.

Key Design Decisions

Decision	Choice	Rationale
Execution model	Hybrid neuron-based routing by capability	Enables user-specific MCP execution (desktop) alongside server-side compute
Gateway placement	Mounted on `apps/api` for v1, isolated enough to extract	Faster to ship; scale-out path preserved
Transport	SSE (server→neuron) + HTTPS POST (neuron→server)	Enterprise-network friendly; WebSocket fails behind many corporate proxies
Auth	JWT from Moku (existing IDP integration); verified in-process via cached JWKS	Reuses existing auth; no per-request Moku round-trip
Capability scoping	Per-advertisement (`any` or `onBehalfOf:userId`); auth token bounds `onBehalfOf`	Prevents user-scope spoofing; one neuron may mix scopes
Queue topology	Per-user queues lazily created, broker-reaped via `x-expires`	Natural per-user ordering; pending-state visibility; bounded growth
Lease semantics	Unacked AMQP messages + DB `assigned` state + lease expiry reaper	Restart-from-scratch on disconnect; no partial-progress preservation
Handler idempotency	Author contract (framework does not provide mid-task checkpointing)	Simpler framework; leverages capability-native idempotency keys
Concurrency	Per-capability advertisement → `basic.qos prefetch_count`	Broker handles backpressure; no cross-capability global cap in v1
Schema ownership	Per-plan catalog, snapshotted with plan; no global capability registry	Enables desktop-unique capabilities; avoids coordination bottleneck
Namespace authorization	Capability type = `{owner}/{package}/{name}` ; gateway enforces identity → namespace table	Prevents capability squatting; silent mis-routing becomes visible registration failure
Wire protocol	`@workflow/neuron-protocol` package shared between gateway and SDK	Single source of truth for message shapes
Neuron SDK	`@workflow/neuron-sdk` ; used by built-in worker too	One implementation, one test surface
Policy integration	Four hooks (register / gather / validate / execute) with NoOp shim shipped first	Locks call sites without blocking on real engine design
Moku boundary	BigBrain owns its schema; Moku retains read for reporting	Coordination at schema-change time, not runtime

Failure Modes

Neuron disconnect mid-task

Broker redelivers unacked message. Reaper flips `assigned` → `pending`. Next consumer binding (same user, any device) re-leases. Handler must be idempotent.

User-scoped neuron offline at dispatch

Task sits in per-user queue (`pending` state in DB, visible in UI). `x-message-ttl` bounds the wait window; on expiry the task DLX's to the planner for re-plan or fail.

Capability collision attempt

Gateway rejects registration at the namespace-authorization check. Visible failure, surfaced to neuron with a specific error code.

Schema mismatch at execution

Neuron SDK pre-validates input before invoking handler; rejects with `schema-mismatch` nack. Output validated on ack; a mismatching output also nacks. Either surfaces through the result processor as a terminal task failure with a structured error.

Policy change mid-workflow

`checkExecution` compares at-plan-time snapshot to current policy. Divergence → task fails with a specific `policy-changed` error that names the rule and the plan-time version. The workflow can be re-planned against current policy if the user retriggers.

Gateway restart

Neurons notice SSE disconnect, attempt `register` via reconnection loop. All in-flight leases were unacked → broker redelivered them → reaper already moved them back to pending. Nothing to recover explicitly; connection re-establishment picks up naturally.

Open Items (Deferred)

- **Real policy engine design** — interface stable; engine design (rule language, classification propagation, obligations) deferred.
- **Device credentials** — v1 uses user-session JWTs; device-specific credentials with per-device revocation are a follow-up.
- **Cross-capability global concurrency cap** — per-capability caps only in v1; add global if a real case emerges.
- **Browser-based neurons** — SDK is node-first; wire protocol stays browser-compatible for future.
- **Per-capability feature tags for version drift** — not in v1; add when needed.